ISyE 3770 HW5

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Problem 5.1.4

Part A:

We observe this problem to resemble a hypergeometric distribution, with a PMF as follows:

$$f_{XY}(x,y) = \frac{\binom{30}{x}\binom{10}{y}\binom{60}{4-x-y}}{\binom{100}{4}}$$

Therefore, we have the following table:

$X \mid Y$	0	1	2	3	4	Total $f_X(x)$
0	0.1244	0.0873	0.02031	0.0018	0.0001	0.234
1	0.2618	0.13542	0.02066	0.00092	0	0.419
2	0.1964	0.0666	0.00499	0	0	0.268
3	0.0621	0.01035	0	0	0	0.073
4	0.0069	0	0	0	0	0.007
Total $f_Y(y)$	0.6516	0.2996	0.0460	0.0028	0.0001	1

Part B:

X	0	1	2	3	4
$f_X(x)$	0.234	0.419	0.268	0.073	0.007

Part C:

$$E[X] =] \sum_{x=0}^{4} x f_X(x) = 0(0.234) + 1(0.419) + 2(0.268) + 3(0.073) + 4(0.007)$$

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0 * 0.234 + 1 * 0.419 + 2 * 0.268 + 3 * 0.073 + 4 * 0.007
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## [1] 1.202
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Part D:

We know the following:

$$P(A|B) = \frac{P(A,B)}{P(B)}$$

Therefore, we have

$$f_{Y|X=3}(y) = \frac{f_{XY}(3,y)}{f_X(3)} = \frac{f_{XY}(3,y)}{0.0725}$$

And so, we arrive at our answer of

$$f_{Y|X=3} = \begin{array}{|c|c|c|c|c|c|}\hline X & 0 & 1 & 2 & 3 & 4 \\ \hline f_X(x) & 0.857 & 0.143 & 0 & 0 & 0 \\ \hline\end{array}$$

Part E:

$$E[Y|X=3] = \sum_{y=0}^{4} y f_Y(y|X=3) = 0(0.857) + 1(0.143) + 0 + 0 + 0$$

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0 * 0.857 + 1 * 0.143 + 0 + 0 + 0
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## [1] 0.143
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Part F:

$$V(Y|X=3) = 0^2(0.857) + 1^2(0.143) - (0.143)^2 = 0.123$$

Part G:

For independence, we know $f_{XY}(x,y) = f_X(x)f_Y(y)$. Therefore, we find the following:

$$f_Y(2) \neq 0$$
$$f_{Y|X=3}(2) = 0, \; as\ previously\ found$$

Therefore, since the previous two equations are not equal, they are not independent.

Problem 5.1.9

$$\int_0^3 \int_0^3 cxydxdy = \int_0^3 (cy\frac{x^2}{2})|_0^3 dy = \int_2^3 \frac{9}{2}cydy = (\frac{9cy^2}{4})|_0^3 = \frac{81c}{4} = 1$$

Therefore, we have $c = \frac{4}{81}$.

Part A:

$$P(X<2, Y<2) = \int_0^3 \int_0^2 \frac{4}{81}xydxdy = \int_0^3 (\frac{4yx^2}{162})|_0^2 dy = (\frac{81y^2}{162})|_0^3 = \frac{4}{9} = 0.4444$$

Part B:

$$P(X<2.5) = \int_0^3 \int_0^{2.5} \frac{4}{81}xydxdy = \int_0^3 (\frac{4yx^2}{162})|_0^{2.5} dy = (\frac{12.5y^2}{162})|_0^3 = \frac{6.25}{9} = 0.694444$$

Part C:

$$P(1<Y<2.5) = \int_1^{2.5} \int_0^3 = \int_1^{2.5} (\frac{4yx^2}{162})|_0^3 dy = (\frac{2y^2}{18})|_1^{2.5} = \frac{7}{12} = 0.5833$$

Part D:

$$P(X>1.8, 1<Y<2.5) = \int_1^{2.5} \int_{1.8}^3 \frac{4}{81}xydxdy = \int_1^{2.5} \frac{32y}{225}dy = \frac{28}{75} = 0.3733$$

Part E:

$$E[X] = \int_0^3 x f_X(x)dx = \int_0^3 x\frac{2x}{9} = (\frac{2x^3}{27})|_0^3 = 2$$

Part F:

$$P(X < 0, Y < 4) = 0, \ \textit{since X is restricted to } 0 < x < 3$$

Part G:

$$f_X(x) = \int_0^3 \frac{4}{81}xydy = (\frac{2xy^2}{81})|_0^3 = \frac{2x}{9}, \ \textit{where } 0 < x < 3$$

Problem 5.4.2

$$\sum_{x=1}^{3}\sum_{y=1}^{3} c(x+y) = \sum_{x=1}^{3} c((x+1)+(x+2)+(x+3)) = \sum_{x=1}^{3} c(3x+6) = 3(1+2+3)+3(6) = c(18+18) = c36 = 1$$

Therefore, we find $c = \frac{1}{36}$.

Next, we calculate Covariance.

$$E[XY] = \sum_{x=1}^{3}\sum_{x=1}^{3} xy(\frac{1}{36})(x+y)$$

For $x = 1$:

$$\begin{aligned}
y = 1: &\quad 1 \cdot 1 \cdot \frac{1}{36}(1+1) = \frac{2}{36} \\
y = 2: &\quad 1 \cdot 2 \cdot \frac{1}{36}(1+2) = \frac{6}{36} \\
y = 3: &\quad 1 \cdot 3 \cdot \frac{1}{36}(1+3) = \frac{12}{36}
\end{aligned}$$

For $x = 2$:

$$\begin{aligned}
y = 1: &\quad 2 \cdot 1 \cdot \frac{1}{36}(2) = \frac{4}{36} \\
y = 2: &\quad 2 \cdot 2 \cdot \frac{1}{36}(2+2) = \frac{16}{36} \\
y = 3: &\quad 2 \cdot 3 \cdot \frac{1}{36}(2+3) = \frac{30}{36}
\end{aligned}$$

For $x = 3$:

$$\begin{aligned}
y = 1: &\quad 3 \cdot 1 \cdot \frac{1}{36}(3+1) = \frac{12}{36} \\
y = 2: &\quad 3 \cdot 2 \cdot \frac{1}{36}(3+2) = \frac{30}{36} \\
y = 3: &\quad 3 \cdot 3 \cdot \frac{1}{36}(3+3) = \frac{54}{36}
\end{aligned}$$

$$E[XY] = \frac{2}{36} + \frac{6}{36} + \frac{12}{36} + \frac{4}{36} + \frac{16}{36} + \frac{30}{36} + \frac{12}{36} + \frac{30}{36} + \frac{54}{36}$$

$$E[XY] = 4.66666$$

$$E[X] = \sum_{x=1}^{3} x f_X(x) = \frac{1}{4} + \frac{2}{3} + \frac{5}{4} = \frac{13}{6}$$

By symmetry, $E[Y] = E[X] = \frac{13}{6}$

$$Cov(X,Y) = 4.66666 - \frac{169}{36} = -0.0277$$

Now, we calculate Correlation:

$$\rho(X,Y) = \frac{-0.0277}{\sqrt{\frac{23}{36} \cdot \frac{23}{36}}} = -0.04347$$

Problem 5.4.6

We first calculate covariance.

$$f_X(x) = \int_0^\infty e^{-x-y} dy = e^{-y} \int_0^\infty e^{-x} dx = e^{-x}$$

$$f_Y(y) = e^{-y}$$

$$E[X] = \int_0^\infty x e^{-x} dx = (-xe^{-x})|_0^\infty + \int_0^\infty e^{-x} dx = 0 + 1 = 1 = E[Y]$$

$$E[XY] = (\int_0^\infty x e^{-x} dx)(\int_0^\infty y e^{-y} dy) = (1)(1) = 1$$

$$Cov(X,Y) = E[XY] - E[X]E[Y] = 1 - (1)(1) = 0$$

Now, we will calculate correlation.

$$Var(X) = E[X^2] - (E[X])^2 = \int_0^\infty x^2 e^{-x} dx - 1^2 = 2 - 1^2 = 1 = Var(Y)$$

$$\rho = \frac{Cov(X,Y)}{\sqrt{Var(X)Var(Y)}} = \frac{0}{1 \cdot 1} = 0$$